Module 02

K8s - kubectl cli commands

DevOps end to end services & solutions

Agenda

- K8s - YAML files
- Kubectl common CLI commands
- Hands-on

K8s – Yaml Files

YAML is a human-readable data serialization language
that is often used for writing configuration files

It's widely used in tools like Ansible,
Kubernetes, and Docker Compose
for orchestrating and automating tasks
across the development lifecycle

Infrastructure as Code (IaC):
YAML facilitates Infrastructure as Code,
allowing DevOps teams to describe
and provision infrastructure components
programmatically

K8s – Yaml Files

POD Service ReplicaSet

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: my-app
spec:
  containers:
    - name: my-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

```yaml
apiVersion: v1
kind: Service
metadata:
  name: example-service
spec:
  selector:
    app: my-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: ClusterIP
```

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: example-replicaset
  labels:
    app: my-app
spec:
  replicas: 3
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
        - name: my-container
          image: nginx:latest
          ports:
            - containerPort: 80
```

K8s – Yaml Files

Example Selector (As Filter) and Labels connection

Deployment configuration:

Deployment

ReplicaSet

Pod

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.7.9
        ports:
        - containerPort: 80
```

Service configuration:

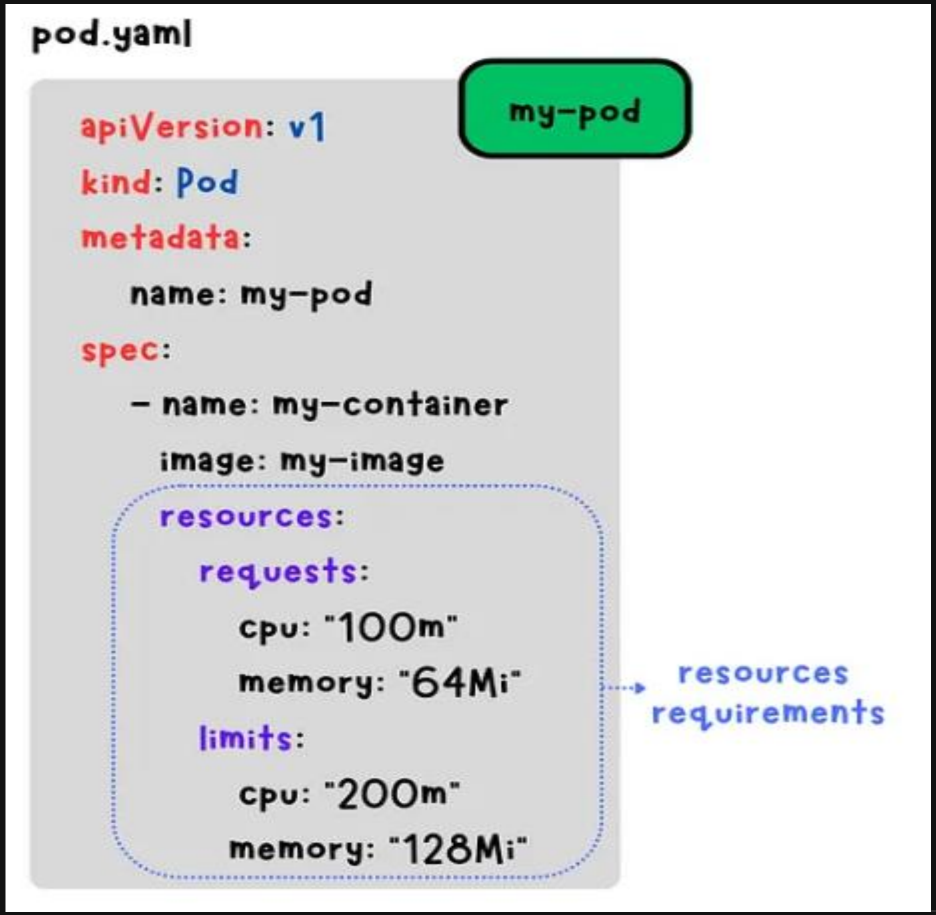
```yaml
kind: Service
apiVersion: v1
metadata:
  name: my-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 9376
```

Ingress configuration:

```yaml
apiVersion: extensions/v1beta1
kind: Ingress
metadata:
  name: my-ingress
  annotations:
    kubernetes.io/ingress.class: nginx
spec:
  rules:
  - host: foo.bar.com
    http:
      paths:
      - path: /foo
        backend:
          serviceName: my-service
          servicePort: 80
      - path: /bar
        backend:
          serviceName: my-other-service
          servicePort: 80
```

K8s – Yaml Files

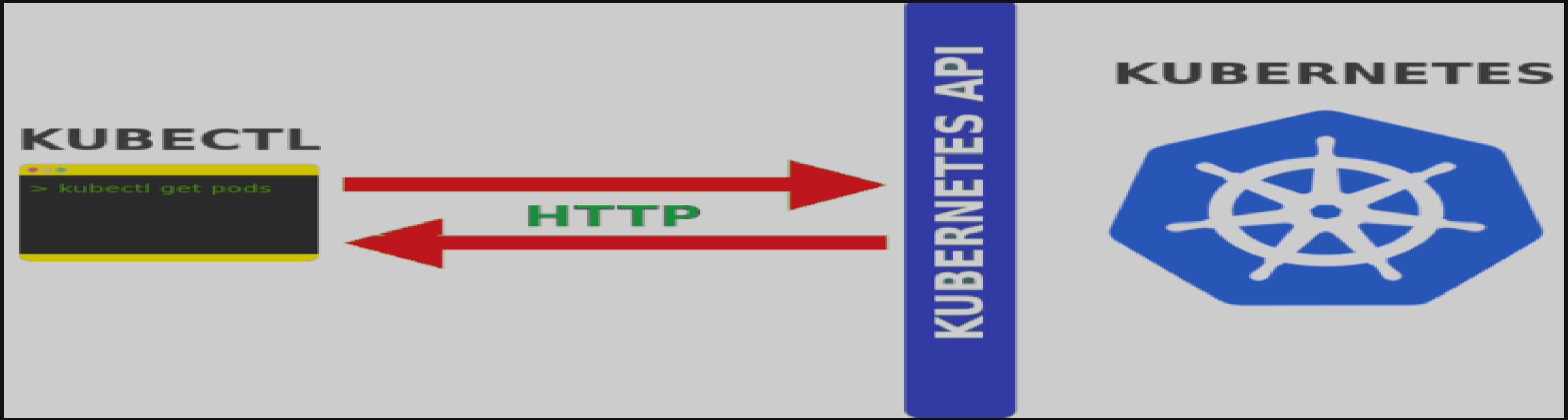
It could be Error Out Of Memory: Kubernetes uses YAML files to specify resource requirements for pods and containers, including CPU and memory resources



K8s – Kubectl common CLI commands

- Kubectl is a command line tool used to run commands against Kubernetes clusters

- It does this by authenticating with the Master Node of your cluster and making API calls to do a variety of management actions



K8s – Kubectl common CLI commands

- kubectl cluster-info
- kubectl version
- kubectl get nodes
- kubectl describe node kind-01-worker
- kubectl get pod
- kubectl get svc
- docker ps
- docker images

K8s – Kubectl common CLI commands

- kubectl cluster-info
- kubectl get pods --all-namespaces
- kubectl describe pod nginx-676b6c5bbc-h2p6m
- kubectl logs nginx-676b6c5bbc-h2p6m
- kubectl logs nginx-676b6c5bbc-h2p6m -c nginx
- kubectl exec -it nginx-676b6c5bbc-h2p6m --/bin/bash
- kubectl get svc
- kubectl describe svc nginx
- kubectl get namespaces
- kubectl config set-context --current --namespace=<>

Hands-On – Lab 02

Getting Started

- Create namespace with name ns-my-first-pod based on exist cluster
- Get the info of cluster
- Get the version of kubectl
- Get the minimal info of nodes
- Get overall info of nodes
- Get the info of namespace (ns)
- What is the internal IP of nodes ?

Questions?

Thank you for participating.

Feel free to reach out:
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